

STRENGTH IN NOTCHED LAMINATES

DEFENCE AND SPACE

Section: 3.4 – STRENGTH OF COMPOSITE PARTS

Chapter: 3404

Title: STRENGTH IN NOTCHED LAMINATES

Revision: 1.1

Date: 11/10/2024

1 INTRODUCTION

1.1 OBJECT, SCOPE AND LIMITATIONS

This chapter presents methodologies to predict the strength of fibrous composite laminates including notches causing stress concentration. In current revision only laminates open holes under by-pass in-plate loads are covered. It is still considered applicable with a moderate level of bending loads. The insights of the proposed methodology is exhaustively described in Ref. 10, and indeed not all the details explained there will be presented in this chapter.

Risk of loss of buckling stability in shear or compression loads is not treated herein.

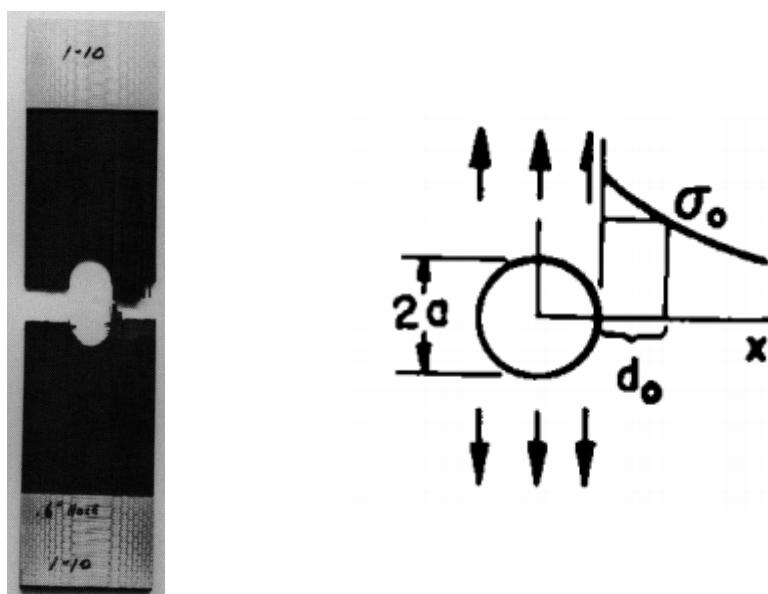


Figure 1. Composite plate with open hole under tension load

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2 STRESS AND STRENGTH OF HOLED LAMINATES

2.1 STRESS DISTRIBUTION AROUND HOLE

By-pass stress in infinite plate

The stress / strain levels at the edge of the hole with the plate subjected to in-plane loads can be calculated according to ESDU85001.

For the more simple case of a notched laminate in tension load, aligned with X-direction, with peak stresses at the transversal section (derived by several authors, from Lekhnitskii's solution), the expected stress concentration factor is:

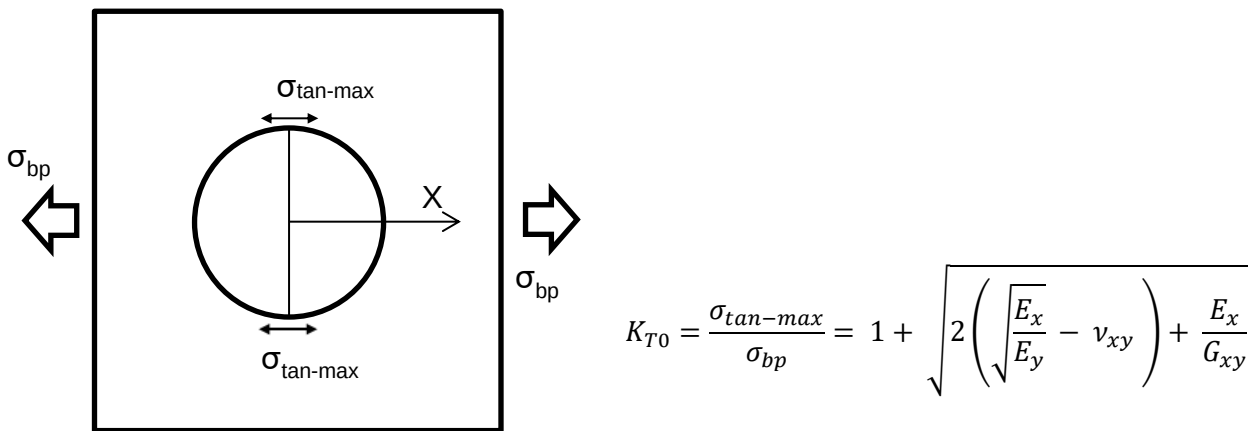


Figure 2. By-pass gross stress and maximum stress around hole

Notice that this calculation does not consider the stress relief due to the filled-hole condition.

Elastic moduli at directions different from X and Y (at an angle θ relative to X / material reference axis) can be calculated using expressions in literature, as Jones' equations 2.97 in page 81.

Finite width correction (d/w)

The following relationship between the stress concentration factors for the infinite plate and a finite width strip, proposed by Tan (expression for circular holes on QI laminates), shall be used:

$$K_{fwc} = K_T / K_{T0} = [2 + (1 - d/w)^3] / [3 \cdot (1 - d/w)]$$

where d is the rivet/hole diameter, w the effective width, K_{T0} the stress concentration factor in an infinite plate and K_T the corrected stress concentration factor.

For any laminate (QI and others), literature proposes the following expression for the finite width correction (FWC) factor:

$$K_{fwc} = \frac{K_T}{K_{T0}} = \left\{ \frac{3(1 - d/w)}{2 + (1 - d/w)^3} + \frac{1}{2} \left(\frac{d}{w} M \right)^6 (K_{T0} - 3) \left[1 - \left(\frac{d}{w} M \right)^2 \right] \right\}^{-1}$$

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where the parameter M is calculated as:

$$M^2 = \frac{\sqrt{1 - 8 \left[\frac{3(1 - d/w)}{2 + (1 - d/w)^3} - 1 \right]} - 1}{2(d/w)^2}$$

For a typical d/w ratios equal to $1/4$ (test method for FHT in Airbus I+D-E-326 standard) and $1/5$ (test methods for OHT, FHT, OHC and FHC in AITM1-0007 and AITM1-0008), the calculated stress/strain in an infinite plate should be multiplied by the factor:

$$K_{fwc}(d/w=1/4) = K_T / K_{T0} = [2 + (1 - 0.25)^3] / [3 \cdot (1 - 0.25)] = 1.0764$$

$$K_{fwc}(d/w=1/5) = K_T / K_{T0} = [2 + (1 - 0.2)^3] / [3 \cdot (1 - 0.2)] = 1.0467$$

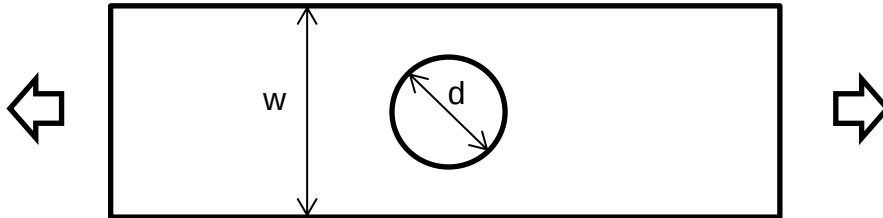
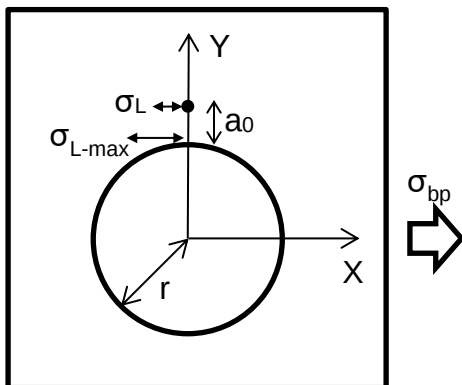


Figure 3. Notched plate with finite width (w)

For holed plates under in-plane shear, the correction proposed by ESDU 75034 (Ref. 11) shall be applied instead.

Stress relaxation near hole edge

As per Lekhnitskii's derivation and formulations of Tan, Whitney and Nuismer, the theoretical stress distribution along the critical cross section of a hole laminate in by-pass load can be calculated with following expressions, where y is the point distance to hole centre, a_0 the distance to the hole edge, r the hole radius and d the hole diameter.



Non-dimensional edge distance:

$$\xi = y / r = (r + a_0) / r = 1 + 2 \cdot a_0 / d$$

Stress concentration:

$$K_{T\xi} = \sigma_L / \sigma_{bp}$$

$$K_{T\xi} = 1 + 0.5 / \xi^2 + 1.5 / \xi^4 - 0.5 \cdot (K_{T0} - 3) \cdot (5 / \xi^6 - 7 / \xi^8)$$

Relaxation factor:

$$K_\xi = K_{T\xi} / K_{T0}$$

Figure 4. Stress at certain distance (a_0) from the hole edge

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Failure criterion

Open-holed laminates can sustain loads beyond the moment when the peak stress at the hole edge reaches the material (plain) strength. For dealing with behaviour, Whitney and Nuismer suggested the concept of “characteristic distance” a_0 , which can be defined as the distance from the hole edge, set at failure level, to the point in the material where the calculated stress matches the material strength.

The characteristic distance (a_0) shall be initially calibrated from the open-hole testing typically performed during material qualification, in specimens made of quasi-isotropic laminates, with hole-diameter to width to ratios equal to 0.2, and hole diameters equal to 6.35 mm (1/4 in). In order to account for the effect of the non-isotropy, an empirical correction shall be then applied to this parameter (details in Ref. 10).

The same allowable strain applicable for checking the plain-strength failure shall be compared with the tangential strain, calculated as discussed in previous paragraphs at that (corrected) characteristic distance, in order to obtain, finally, the static margin.

3 STRENGTH OF LAMINATES WITH OTHER TYPES OF NOTCHES

Reserved for future revisions.

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4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a, a_0	mm	Distance, or characteristic distance, from the hole edge to a point in the material, in radial direction
b_p	-	By-pass (gross) stress, strain or load, as subscript
d	mm	Diameter of hole
E_x	MPa	Apparent elastic modulus of the laminate in X direction
E_y	MPa	Apparent elastic modulus of the laminate in Y direction
G_{xy}	MPa	Apparent shear elastic modulus of the laminate in its plane
ν_{xy}	-	Poisson's ratio of the laminate
N_x, N_y, N_{xy}	N/mm	In-plane forces per unit width
K	-	Relaxation factor, amplification factor, or knock down factor (from context)
K_T	-	Stress concentration factor
K_{T0}	-	Stress concentration factor in the infinite plate
r	mm	Hole radius
t	mm	Thickness of the laminate
w	mm	Width of the plate
ε	-	Strain
σ	MPa	Stress
x, y	-	Directions of the reference axis system of the laminate

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[R.M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 2	[S. G. Lekhnitskii]	1968	Anisotropic Plates
Ref. 3	ESDU85001	1985	ELASTIC STRESS AND STRAIN DISTRIBUTIONS AROUND CIRCULAR HOLES IN INFINITE PLATES OF ORTHOTROPIC MATERIAL (Applicable to Fibre Reinforced Composites)
Ref. 4	[J. M. Whitney, R. J. Nuismer]	1974	Stress Fracture Criteria for Laminated Composites Containing Stress Concentrations

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Ref. 5	[S.C. Tan]	1986	Laminated Composites Containing an Elliptical Opening. I. Approximate Stress Analyses and Fracture Models
Ref. 6	[S.C. Tan]	1986	Laminated Composites Containing an Elliptical Opening. II. Experiment and Model Modification
Ref. 7	[S.C. Tan]	1988	Finite-Width Correction Factors for Anisotropic Plate Containing a Central Opening
Ref. 8	[S.C. Tan]	1994	Tan SC. Stress concentrations in laminated composites
Ref. 9	[C. Furtado et al]	2017	Prediction of size effects in open-hole laminates using only the Young's modulus, the strength, and the R-curve of the 0° ply
Ref. 10	TE-T-NT-240001	Is. B	METHOD FOR CHECKING THE STRENGTH OF COMPOSITE LAMINATES WITH AN OPEN HOLE
Ref. 11	ESDU 75034	1975	INITIAL BUCKLING STRESS, MAXIMUM DIRECT STRESS AND SHEAR STRAIN OF SQUARE PLATES IN SHEAR WITH CENTRAL CIRCULAR HOLES

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 12/04/2024	G. Baños	First issue	All pages
1.1 11/10/2024	G. Baños	Alignment with the update of the open-hole method definition report Ref. 10	Pages 1, 3, 4, 6

Table 1. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods & Optimization	G. Baños 11/10/2024	-
Structural Analysis	-	F. Glock 07/01/2025

Table 2. *Approval signatures table for last revision of this chapter*